HAna Note

1 Sup Inf & Set || Nested Intervals || Sequence & Bolzano Weierstrass

1.1 Supremum and Infimum & Set

Completeness Axiom: If $\exists$ UP $\Rightarrow$ LUB (sup) ; If $\exists$ LB $\Rightarrow$ GLB (inf). **Archimedean Property**: $\forall x \in \mathbb{R}^+ \Rightarrow \exists n \in \mathbb{N}\ s.t.\ n > x$.

Countable Set: A set E is countable if $\exists$ bijection $f : \mathbb{N} \to E$. ps: $[0,1]$ is uncountable.

· **Property**: Let $(E_n)_{n\in\mathbb{N}}$ be countable sets. $\Rightarrow$ $^1 \bigcup E_n$ is countable. (Can infinite) $^2 \prod E_n$ is countable. (Finite) $^3 [0,1]$ uncountable.

$^\oplus$ **Dense Set**: Set A is dense in $\mathbb{R}$ iff $\forall x \in \mathbb{R}, \forall \varepsilon > 0\ \exists a \in A$ s.t. $|x - a| < \varepsilon$ e.g.$^\oplus$ $\mathbb{Q}$ is dense in $\mathbb{R}$

$^\oplus$ **Power Set** $\mathcal{P}$: $P(E)$ is the set of all subsets of E. ps: $\mathcal{P}(\mathbb{R})$ is the set of all subsets of $\mathbb{R}$.

Property of Sup , Inf: $\inf(-f_n) = -\sup(f_n)$ || $\limsup(-f_n) = -\liminf(f_n)$ || If g is fixed, $\sup(f_n + g) = \sup(f_n) + g$ ps: 对 inf, lim sup, lim inf 也成立

1.2 Nested Intervals

Def of $\lambda(I)$: Let I be an interval. $\lambda(I) := \sup I - \inf I$ || $\lambda((a,b)) = \lambda([a,b)) = \lambda((a,b]) = \lambda([a,b]) = b - a$.

Nested Intervals: Sequence $(I_n)_{n\in\mathbb{N}}$ is nested if $I_n \supseteq I_{n+1}$ ($\downarrow$) for all $n \in \mathbb{N}$.

Nested Interval Theorem: If $(I_n)_{n\in\mathbb{N}}$ is nested *closed bounded* interval sequence $\Rightarrow E := \bigcap I_n \neq \emptyset$.

· **property**: ^{1}If $\lim \lambda(I_n) = 0 \Rightarrow E = a$ $^{2\oplus}$ If $I_n = [a_n, b_n]$ $\Rightarrow$ $E = [\sup a_n, \inf b_n]$

1.3 Sequence, Cauchy & Bolzano Weierstrass Theorem

Useful Seq Limit: $\lim \frac{1}{n^p} = 0, p > 1$; $\lim a^n = 0, |a| < 1$; $\lim n^{\frac{1}{n}} = 1$; $\lim a^{\frac{1}{n}} = 1, a > 0$

Operation in Seq Limit: *Squeeze Theorem* ; *Monotone Convergence Theorem (MCT)* ; $\lim \alpha x_n = \alpha \lim x_n$

· $\lim(x_n + y_n) = \lim x_n + \lim y_n$; $\lim x_n \cdot y_n = \lim x_n \cdot \lim y_n$; $\lim \frac{x_n}{y_m} = \frac{\lim x_n}{\lim y_n}, y_n \neq 0$

Comparison Theorem (For Seq): If $\lim x_n, y_n$ exist, and $x_n \leq y_n$ for all $n \geq N$ then $\lim x_n \leq \lim y_n$.

Cauchy Sequence: Sequence $(a_n)_{n\in\mathbb{N}}$ is Cauchy iff $\forall \varepsilon > 0, \exists N \in \mathbb{N}\ s.t.\ |a_n - a_m| < \varepsilon$ for all $m, n \geq N$. || **Convergent $\Leftrightarrow$ Cauchy**

Bolzano-Weierstrass Theorem: Every *bounded sequence* has a *convergent subsequence*.

Def of limit superior and inferior: $\limsup a_n := \lim(sup_{k\geq n} a_k)$ || $\liminf a_n := \lim(inf_{k\geq n} a_k)$

· **Property**: If $(a_n)_{n\in\mathbb{N}}$ bounded. $\Rightarrow$ $^{\oplus 1}\exists(a_{n_k}) \to \limsup a_n$ Hint:$\exists a_{n_k}, \sup_{j\geq k} a_j - \frac{1}{k} \leq a_{n_k} \leq \sup_{j\geq k} a_j, \sup_{j\geq n_k} a_j \leq \sup_{j\geq k} a_j$ $^{\oplus 2}\exists(a_{n_k}) \to \liminf a_n$
$^{\oplus 3}\forall a_{n_k} \to a$ $\liminf a_n \leq a \leq \limsup a_n$

Subsequence Theorem: In each case, the subsequence can be taken to be *monotonic*.

1. $\exists a \in \mathbb{R}\ s.t.\ \forall \varepsilon > 0$, there are infinitely many $n \in \mathbb{N}$ $s.t.\ |x_n - a| < \varepsilon$ ($\exists$ 聚点 a) $\Leftrightarrow$ $\exists$ subsequence of (x_n) $s.t.$ $\lim x_{n_k} = a$
2. $(x_n)_{n\in\mathbb{N}}$ is not bounded above (below) $\Leftrightarrow$ $\exists$ subsequence of x_n $s.t.$ $\lim x_{n_k} = \infty$ $(-\infty)$
3. If $\lim x_n = L \Rightarrow \forall x_{n_k}$, $\lim x_{n_k} = L$

2 Series

Cauchy Criterion for Series: $\sum_{k=1}^{\infty} a_k$ converges $\Leftrightarrow$ $\forall \varepsilon > 0, \exists N \in \mathbb{N}\ s.t.\ \forall n \geq m \geq N \mid \sum_{k=m+1}^{n} a_k| = |a_{m+1} + \cdots + a_n| < \varepsilon$.

p-Test: $\sum \frac{1}{n^p}$ convergent iff $p > 1$ **Root Test** **Ratio Test**$_{|n|^{1/n}, |n+1/n|}$ **Geometric Series**: $\sum p^n$ converges iff $|p| < 1$

Divergence Test: If $\lim a_n \neq 0 \Rightarrow \sum a_n$ diverges. (If $\sum a_n$ convergent $\Rightarrow \lim a_n = 0$.)

Comparison Test: If $0 < a_n < b_n, \sum b_n$ convergent $\Rightarrow \sum a_n$ also ; $\sum a_n$ divergent $\Rightarrow \sum b_n$ also.

· **Limit Comparison Test**: If $a_n, b_n \geq 0, \lim \frac{a_n}{b_n} = L > 0$ $\Rightarrow$ If $L \in (0, \infty)$: $\sum a_n$ convergent iff $\sum b_n$ also.
If $L = 0 : \sum b_n$ convergent $\Rightarrow \sum a_n$ also. $L = \infty : \sum b_n$ divergent $\Rightarrow \sum a_n$ also.

Integral Test: Let $f : [1, \infty) \to \mathbb{R}$ is 非负递减, $a_n = f(n)$. Then $\sum a_n$ converges iff $\int_1^{\infty} f(x)dx < \infty$.

Absolutely Convergence: $\sum a_n$ convergent absolutely iff $\sum |a_n|$ convergent. **If convergent abs $\Rightarrow$ convergent.**

Alternating Series Test: If a_n decreasing, $a_n \geq 0, \lim a_n = 0$. Then $\sum(-1)^{n-1} a_n$ convergent.

Cauchy's Condensation Test: If $a_n \geq 0, a_n$ decreasing, $\Rightarrow$ $[\sum a_n convergent \Leftrightarrow \sum 2^n a_{2^n}\ also\ ]$

Riemann Rearrangement Theorem: If $\sum_{k=1}^{\infty} a_k$ is conditionally convergent. $\Rightarrow$ $\forall s \in \mathbb{R}, \exists$ bijection $\sigma : \mathbb{N} \to \mathbb{N}$ s.t. $\sum_{n=1}^{\infty} a_{\sigma(n)} = s$.

Rearrangement for Abs Series: If $\sum_{n=1}^{\infty} a_n$ is absolutely convergent. $\Rightarrow$ $\forall$ bijection $\sigma : \mathbb{N} \to \mathbb{N}$, $\sum_{n=1}^{\infty} a_{\sigma(n)} = \sum_{n=1}^{\infty} a_n$.

3 Continuous & Differentiable Functions Properties

Def of lim of func: **1**.$\forall \varepsilon > 0, \exists \delta > 0\ s.t. 0 < |x - x_0| < \delta, \Rightarrow |f(x) - L| < \varepsilon$. **2**.$\forall x_n \neq x_0\ s.t. \lim(x_n) = x_0, then \lim f(x_n) = L$.

Def of cont of func: **1**.$\forall \varepsilon > 0, \exists \delta > 0\ s.t.\ \forall x, |x - x_0| < \delta \Rightarrow |f(x) - f(x_0)| < \varepsilon$. **2**.$\forall (x_n)_{n\in\mathbb{N}}, if\ \lim x_n = x_0 \Rightarrow \lim f(x_n) = f(x_0)$

Def of diffi: **1**.$f'(a) = \lim_{x\to a} \frac{f(x)-f(a)}{x-a}$ **2**.$f'(a) = \lim_{h\to 0} \frac{f(a+h)-f(a)}{h}$.

The Extreme Value Theorem (EVT): If $f : [a,b] \to \mathbb{R}$ is continuous, then f has a max and min value.

Intermediate Value Theorem (IVT): $f : [a,b] \to \mathbb{R}$ is continuo,$f(a) \neq f(b), y \in [f(a), f(b)]$.$\Rightarrow \exists c \in (a,b)\ s.t. f(c) = y$

IVT (for deriv) / (Darboux Theorem): If f differentiable in $I \Rightarrow \forall m \in (f'(a), f'(b))$ we have: $\exists c$ s.t. $f'(c) = m$

Rolle's Theorem: If f continuous on $[a,b]$, differentiable on (a,b) and $f(a) = f(b)$ Then: $\exists c \in (a,b)$ s.t. $f'(c) = 0$

Mean Value Theorem (MVT): Let f, g continuous on $[a,b]$ and differentiable on (a,b): ps: a=b 时依然成立

· **1**.$\exists c \in (a,b)$ s.t. $\frac{f(b)-f(a)}{b-a} = f'(c)$ **2**. $\exists c \in (a,b)$ s.t. $\frac{f(b)-f(a)}{g(b)-g(a)} = \frac{f'(c)}{g'(c)}$

Theorem: If f is 1-1 and continuous $\Rightarrow$ f is strictly monotone. and f^{-1} is continuous and strictly monotone.

Inverse Function Theorem: Let f differentiable in I (open interval), and $\forall x, f'(x) \neq 0$:

· **1.** $f(I)$ is open interval. **2.** f is invertible. **3.** f^{-1} is differentiable and $(f^{-1})'(y) = \frac{1}{f'(f^{-1}(y))}$

Second Deriv Test: f twice continuously differentiable in I,$f'(x_0) = 0$: **1.** $f''(x_0) > 0$ (< 0) local min (max) point.

4 Pointwise & Uniformly Convergence

4.1 Pointwise Convergence

Pointwise Convergence: Let $(f_n)_{n\in\mathbb{N}}$ be a sequence of functions. $f_n : E \to \mathbb{R}$ converge pointwise to $f : E \to \mathbb{R}$ if: 不加说明默认 pointwise

1. **Version 1**: $\forall x \in E, \forall \varepsilon > 0, \exists N_{\varepsilon,x}\ s.t.\ \forall n > N\ |f_n(x) - f(x)| < \varepsilon$
2. **Version 2**: $\forall x \in E, \lim f_n(x)$ exists and $\lim f_n(x) = f(x)$

Pointwise Convergence NOT Preserve: ^{1}*continuity*, ^{2}*differentiability*, $^3(\lim f_n)' \neq \lim f_n'$, $^4 \int \lim f_n \neq \lim \int f_n$

· 对应反例: $f_n : [0,1] \to \mathbb{R} := x^n$ $\quad {}^2 f_n : (-1,1) \to \mathbb{R} := \sqrt{x^2 + \frac{1}{n}}$ $\quad {}^3 f_n : (-1,1) \to \mathbb{R} := \frac{\sin(nx)}{n}$ $\quad {}^4 f_n : (0,1] \to \mathbb{R} := \begin{cases} n, 0 < x \leq \frac{1}{n} \\ 0, \frac{1}{n} < x \leq 1 \end{cases}$

4.2 Uniformly Convergence

Graph: $\Gamma(f) := \{(x, f(x)) | x \in E\}$ ps:$f : E \to \mathbb{R}$

Vertical Neighbourhood $\mathcal{N}(f, \varepsilon) := \{(x, y) | x \in E, |y - f(x)| < \varepsilon\}$ ps:$f : E \to \mathbb{R}$

Uniformly Convergence: $f_n : E \to \mathbb{R}$ converge uniformly iff: || **Uniformly ⇒ Pointwise**

1. **Version 1**: $\forall \varepsilon > 0, \exists N_\varepsilon\ s.t.\ \forall n > N, \forall x \in E, |f_n(x) - f(x)| < \varepsilon$
2. **Version 2**: $\lim \left[\sup_{x\in E} |f_n(x) - f(x)|\right] = 0$
3. **Version 3 | Uniformly Cauchy**: $\forall \varepsilon > 0, \exists N_\varepsilon \in \mathbb{N}\ s.t.\ \forall x \in E, m, n > N\ |f_n(x) - f_m(x)| < \varepsilon$ || **Converge Unif ⇔ Unif Cauchy**
4. **Version 4 | Vertical Neighbourhood Test**: $\forall \varepsilon > 0, \exists N \in \mathbb{N}\ s.t.\ \forall n > N, \Gamma(f_n) \subset \mathcal{N}(f, \varepsilon)$

Sequence Test: If $f_n : E \to \mathbb{R}$ converge uniformly to f. $\Rightarrow$ If $(x_n)_{n\in\mathbb{N}} \in E, \lim |f_n(x_n) - f(x_n)| = 0$ 不能当作 $\forall x_n$

Properties of Uniformly Convergent: Let $f_n, f, g : I \to \mathbb{R}$ Then: NOT preserve: differentiable.

1. **Continuous**: If $^1 \lim f_n(x) = f(x)$ uniformly, $^2 \forall f_n$ continuous at x_0 $\Rightarrow$ f continuous at x_0. $\Downarrow \forall x \in I, (\lim f_n)'(x) = \lim f_n'(x)$
2. **Termwise differentiation**: If $^1 I$ is *bounded open*; $^2 \forall f_n$ diff, $\lim f_n' = g$ unif; $\underline{^3 \exists x_0 \in I\ s.t.\ f_n(x_0) \text{ converges.}}$ $\Rightarrow$ $\underline{\lim f_n = f \text{ unif}}$; f' diff and $f' = g$.
3. **Uniform Convergence Theorem**: If $^1 \lim f_n(x) = f(x)$ *uniformly* on $E \in \mathcal{M}, m(E) < \infty$, $^2 f_n \in \mathcal{L}^1(\mathbb{R})$ $\Rightarrow$ $^1 f \in \mathcal{L}^1(E)$ $^2 \lim \int_E f_n(x)dx = \int_E f dx$
4. $^\oplus$ If $f_n : [a,b] \to \mathbb{R}, g_n : [a,b] \to \mathbb{R}$ converge uniformly. $\Rightarrow$ $\frac{f_n}{g_n}$ converge uniformly also. ps:(a,b) 不行 Hint: $f_n g - f g_n = f_n g - fg + fg - f g_n$

Def of Series of Functions: Let $f_n : E \to \mathbb{R}$; Def partial sum $s_n(x) := \sum_{k=1}^n f_k(x)$

1. $\sum f_n$ converge pointwise iff s_n converge pointwise. **2**. $\sum f_n$ converge uniformly iff s_n converge uniformly. **3**.$\sum f_n$ converge absolutely (pointwise) iff $\sum |f_n|$ converge (pointwise).

Weierstrass M-Test: $f_n : E \to \mathbb{R}, \exists (M_n)_{n\in\mathbb{N}} \geq 0$ s.t. $^1 \forall x \in E, \forall n \in \mathbb{N} : |f_n| < M_n$; $^2 \sum M_n$ converges. $\Rightarrow$ $\sum f_n$ converge absolutely & Uniformly.

5 Power Series

5.1 Power Series, Radius of Convergence, Continuity and Differentiability

Power Series: A series from $\sum_{n=0}^\infty a_n(x-c)^n$ **Center**: c. **Coefficients**: a_n.

Radius of Convergence (R): For a power series $\sum a_n(x-c)^n$, there are three situations for it's radius of convergence:

1. $R = 0$ if $\sum a_n(x-c)^n$ converges only $x = c$.
2. $R = \infty$ if $\sum a_n(x-c)^n$ converges absolutely $\forall x \in \mathbb{R}$.
3. $\exists R > 0$ s.t. $\sum a_n(x-c)^n$ converges absolutely if $|x - c| < R$; diverges if $|x - c| > R$.
4. **All-In-One**: $R = \sup\{r \geq 0 : (a_n r^n)_{n\in\mathbb{N}}\ \underline{is\ bounded}\}$

Properties | In Radius of Convergence: Assume $f(x) = \sum a_n(x-c)^n$ with $R > 0$.

1. **Continuous**: $f(x)$ continuous on $(c - R, c + R)$.
2. **Convergence**: $\forall 0 < r < R$, $f(x)$ converges absolutely & uniformly on $[c - r, c + r]$.
3. **Preserve R**: $\forall (a_n)_{n\in\mathbb{N}}$, we have: $\sum_{n=0}^\infty a_n(x-c)^n$ and $\sum_{n=1}^\infty n a_n (x-c)^{n-1}$ have same R. ps: 提供了一种由 $n-1 \leftrightarrow n$ 的方法
4. **Termwise differentiation**: $f(x)$ is differentiable on $(c - R, c + R)$ || $^1 f'(x) = \sum_{n=1}^\infty n a_n (x-c)^{n-1}, x \in (c - R, c + R)$
   $^2 \forall 0 < r < R$, $f'(x)$ converges absolutely and uniformly on $[c - r, c + r]$. ps: 即可以对里面求导
5. **Coefficients & Diff**: $^1 f(x)$ is *infinitely differentiable* on $(c - R, c + R)$ || $^2 a_k = \frac{f^{(k)}(c)}{k!}$
6. **Uniqueness**: If $\sum a_n(x-c)^n = \sum b_n(x-c)^n$ for $\forall x \in \mathbb{R}, |x - c| < r_0$ $\Rightarrow$ $a_n = b_n$

5.2 Exponential Function & Real Analytic Function

Exponential Function: $E(x) := \sum_{n=0}^\infty \frac{x^n}{n!}$ with $R = \infty$ ps: $E(x) = \lim_{n\to\infty}(1 + \frac{x}{n})^n$ Hint: 证 pointwise 考虑 ln,as $ln(1 + \frac{x}{n}) = \int_1^{1+\frac{x}{n}} \frac{1}{t} dt$, 缩放,squeeze theorem.|uniformly 考虑 unif cont

Real Analytic Function: $f : I \to \mathbb{R}$ is real analytic iff $\exists (a_n)_{n\in\mathbb{N}}$ s.t. $f(x) = \sum_{n=0}^\infty a_n(x-c)^n$ for $\forall x \in I$ i.e. Series converges pointwise on I.

Taylor Series: Given ^{1}*Open Interval* I, ^{2}*infinitely differentiable* $f : I \to \mathbb{R}$. $\Rightarrow$ Taylor Series of f at $c \in I := \sum_{n=0}^\infty \frac{f^{(n)}(c)}{n!}(x-c)^n$

· Not all infinitely differentiable functions are real analytic: e.g.$f(x) = \{e^{-1/x^2}, x > 0; 0, x < 0$

Taylor's Theorem: If $f : (a,b) \to \mathbb{R}$ is differentiable function at $x_0 \in (a,b)$:

1. $P_{n,f,x_0}(x) := f(x_0) + \sum_{k=1}^n \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k$ (Taylor's polynomial of degree n at x_0)
2. If $f^{(n+1)}$ exists on (a,b), then: $\exists \xi \in (x, x_0)$ s.t. $f(x) = P_{n,f,x_0}(x) + \frac{f^{(n+1)}(\xi)}{(n+1)!}(x - x_0)^{n+1}$
3. (Lagrange) Remainder $R_{n,f,x_0} := f(x) - P_{n,f,x_0}$. || $f : I \to \mathbb{R}$ is real analytic iff $\lim R_{n,f,x_0} = 0$ for $\forall x \in I$

6 Open & Closed Sets

Open Set: Set E is open iff $\forall x \in E, \exists r > 0,\ s.t. (x-r, x+r) \subseteq E$

1. **$\cup$ of open sets**: (E_n) are open sets (Infinite). $\Rightarrow$ $\cup E_n$ is open.
2. **$\cap$ of open sets**: (E_n) are open sets (Finite). $\Rightarrow$ $\cap E_n$ is open.
3. **Find Open Set**:
   If $f: \mathbb{R} \to \mathbb{R}$ continuous $\Rightarrow$ $\{x \in \mathbb{R} : f(x) > \lambda$ (or $< \lambda$) $\}$ is open.
4. **Structure of open sets**: $\Downarrow^{\ominus}$ can be disjoint Hint: $I'_k = \cup I_n \setminus \cup_{i=1}^{k} I'_i$
   E is a nonempty open set $\Leftrightarrow$ $E = \cup I_n$, I_n is open interval. (Unique)

Closed Set: Set $F \subseteq \mathbb{R}$ is closed iff $\mathbb{R} \setminus F$ is open.

1. **$\cup$ of closed sets**: (F_n) are closed sets (Finite). $\Rightarrow$ $\cup F_n$ is closed.
2. **$\cap$ of closed sets**: (F_n) are closed sets (Infinite). $\Rightarrow$ $\cap F_n$ is closed.
3. **Find Closed Set**:
   If $f: \mathbb{R} \to \mathbb{R}$ continuous $\Rightarrow$ $\{x \in \mathbb{R} : f(x) \le \lambda$ (or $\ge \lambda$) (or $= \lambda$) $\}$ is closed.
4. **Sequentially Closed**:
   If F is closed set, $\Leftrightarrow$ $\forall (x_n)_{n\in\mathbb{N}} \in F$ convergent. $x = \lim x_n \in F$

Open Set $\to \cup$ Closed Set: Nonempty open set $E \to$ union of *disjoint closed* intervals. Hint:^{1}Structure of open set. ^{2} each open interval $(a,b) = \cup[a+\frac{1}{n}, b-\frac{1}{n}]$. ^{3} closed set→disjoint.

Closed Set $\to \cap$ Open Set: Closed set $F \to$ intersection of *open* sets. e.g. $[0,1] = \cap(0-\frac{1}{n}, 1+\frac{1}{n})$

Unbounded$\to \cup$ Bounded: Any closed / open set $F \to$ countable union of *bounded* closed / open sets. (Hint: $K_n = F \cap [-n,n]$ $K_n = F \cap (-n,n)$)

Both Open & Closed: Only $\emptyset$ and $\mathbb{R}$ are both open and closed.

Some Properties of Sets:

1. **De Morgan's Laws**: $(A\cap B)^C = A^C \cup B^C$ $(A\cup B)^C = A^C \cap B^C$ $A\cap(B\cup C) = (A\cap B)\cup(A\cap C)$ $A\cup(B\cap C) = (A\cup B)\cap(A\cup C)$
2. $^{\oplus} E \setminus F = F^C \cap E$ is open (i.e. open - closed = open) || $F \setminus E = E^C \cap F$ is open (i.e. closed - open = closed)

7 Lebesgue Outer Measure || Lebesgue Measurable || Lebesgue Measure

ps: **Countably Additivity**: $(E_n)_{n\in\mathbb{N}}$ disjoint sets, m be measure. $\Rightarrow$ Countably Additive if: $m(\cup_n E_n) = \sum m(E_n)$

7.1 Lebesgue Outer Measure

Lebesgue Outer Measure (m^*): Can interpret m^* as $m^*: \mathcal{P}(\mathbb{R}) \to [0,\infty)$ ps: 本质是寻找到最小的区间可以覆盖 E, 用前面 λ 的定义来定义"长度"

1. **Version 1|Open Intervals**: $m^*(E) := \inf\{\sum_{n=1}^{\infty} \lambda(I_n) \mid E \subseteq \cup_{n=1}^{\infty} I_n, I_n \textit{ is a open interval}\}$
   ps: $\forall \varepsilon > 0, \exists I_n$ open interval s.t. $E \subseteq \cup I_n$ and $m^*(E) < \sum \lambda(I_n) < m^*(E) + \varepsilon$
2. **Version 2|Open Sets**: $m^*(E) = \inf\{\lambda(U), E \subseteq U, U \textit{ is open set}\}$ ps:$\lambda(U) = \sum \lambda(I_n)$
   ps: $\forall \varepsilon > 0, \exists U$ open set s.t. $E \subseteq U$ and $m^*(E) < m^*(U) < m^*(E) + \varepsilon$ or $m^*(E) < \lambda(U) < m^*(E) + \varepsilon$
   · **Remark**: If we cannot find $(I_n)_{n\in\mathbb{N}}$ s.t. $\sum \lambda(I_n) < \infty$ $\Rightarrow$ $m^*(E) = \infty$.

· **Basic 3 Properties**: Let Set E, F $s.t.$ $E \subseteq F \subset \mathbb{R}$.

1. **Null Empty Set**: $m^*(\emptyset) = 0$. || Null Set E: $m^*(E) = 0$.
2. **Monotonicity**: $m^*(E) \le m^*(F)$.
3. **Countable Subadditivity**: $m^*(\cup_{n=1}^{\infty} E_n) \le \sum_{n=1}^{\infty} m^*(E_n)$.

Useful Properties|By Def of inf: Let E be a set, $m^*(E) < \infty$
Then $\forall \varepsilon > 0, \forall \delta > 0$: $\exists (I_n)_{n\in\mathbb{N}}$, I_n open intervals s.t.: $^{\oplus}$
$^1 E \subseteq \cup I_n$ | $^2 m^*(E) \le \sum \lambda(I_n) < m^*(E) + \varepsilon$ | $^3 I_n \cap E \ne \emptyset, \forall n$ | $^4 \lambda(I_n) < \delta, \forall n$
$^5 I_n \cap E \not\subseteq I_k, \forall n \ne k$. ← 无覆盖项 无无关项↑ ps: 也使用与是 open set 的.

· **More Properties**:

1. **Interval**: If $J \subseteq \mathbb{R}$ is interval. $m^*(J) = \lambda(J)$.
2. **Countable**: Any Countable set E $\Rightarrow$ $m^*(E) = 0$ $\exists m^*(E) = 0, E$ uncountable (e.g.Cantor Set (Closed Set))
3. $^{\oplus}$ **Further|$m^{**}(E)$**: $m^*(E) = m^{**}(E) := \inf\{\sum \lambda(I_n) \mid E \subseteq \cup I_n, I_n \textit{ interval}\}$ || $^{\oplus} m^*(E) \le \sum \lambda(I_n)$, if $E = \cup I_n$, I_n intervals.
4. **Translation Invariance**: $m^*(E) = m^*(x+E)$

Def of dist(E, F): $dist(E,F) := \inf\{|x-y| : x \in E, y \in F\}$

Lemma: $^1 E, F$ are nonempty *closed* sets; $^2 E \cap F = \emptyset$; $^3 E$ *bounded*. $\Rightarrow$ $dist(E,F) > 0$

Countable Additivity|Outer Measure:

1. **Basic**: If $dist(E,F) > 0$ $\Rightarrow$ $m^*(E \cup F) = m^*(E) + m^*(F)$.
2. **Closed|Open|Interval Sets**: If $(K_n)_{n\in\mathbb{N}}$ is *disjoint*, (Closed (bdd) | Open | Interval), sets $\Rightarrow$ $m^*(\cup_n^{\infty} K_n) = \sum_{n=1}^{\infty} m^*(K_n)$

7.2 Lebesgue Measurable

Lebesgue Measurable:

1. **Version 1**: A set $E \subseteq \mathbb{R}$ is *Lebesgue measurable* if $\forall \varepsilon > 0, \exists$ open set $U \subseteq \mathbb{R}$ s.t. $E \subseteq U$ and $m^*(U \setminus E) < \varepsilon$ ps: 被 open set 从外"恰好"包住
2. **Version 2**: A set $E \subseteq \mathbb{R}$ is *Lebesgue measurable* if $\forall \varepsilon > 0, \exists$ closed set $U \subseteq \mathbb{R}$ s.t. $U \subseteq E$ and $m^*(E \setminus U) < \varepsilon$ ps: 被 closed set 从内"恰好"包住

· $\mathcal{M}$:= collection of all Lebesgue measurable sets of $\mathbb{R}$.

Property of Lebesgue Measurable Sets:

1. **Closed Set|Open Set|Interval**: Any closed set / open set / interval is Lebesgue measurable.
2. **Union $\cup$|$^{\oplus}$ Intersection $\cap$**: If $(E_n)_{n\in\mathbb{N}}$ is *Lebesgue Measurable* $\Rightarrow$ Also: $^1 E := \cup_{n=1}^{\infty} E_n$ $^{\oplus 2} E := \cap_{n=1}^{\infty} E_n$. Hint: $(\cap E_n)^C = \cup E_n^C$
3. **Complement E^C**: if $E \subseteq \mathbb{R}$ is *Lebesgue Measurable* $\Rightarrow$ $\mathbb{R} \setminus E$ is also *Lebesgue Measurable*.
4. **Measure Zero Set**: $m^*(E) = 0$ $\Leftrightarrow$ E Lebesgue measurable.
5. **Translation Invariance**: $m(E) = m(x+E)$ || If E is Lebesgue Measurable $\Rightarrow$ $x + E$ also.
6. **Find Lebesgue Measurable Set**: A simple function s $\Rightarrow$ $\{x \in \mathbb{R}, s(x) > \lambda(< \lambda \textit{ or } = \lambda)\}$ is Lebesgue Measurable.

7.3 Lebesgue Measure

Lebesgue Measure: *Lebesgue Measure* is a function $m : \mathcal{M} \to [0, \infty)$ s.t. $m(E) := m^*(E), \forall m \in \mathcal{M}$ ps: 原来是 $m^* : \mathcal{P}(\mathbb{R}) \to [0, \infty)$, 现在是 $m : \mathcal{M} \to [0, \infty)$

· **Property**: ⋆ **All properties of Lebesgue Outer Measure hold for Lebesgue Measure.** ⋆

Countable Additivity for Measure: If $(E_n)_{n\in\mathbb{N}}$ *disjoint* & *Lebesgue Measurable.* $\Rightarrow$ $m(\cup_{n=1}^{\infty} E_n) = \sum_{n=1}^{\infty} m(E_n)$

Monotone Convergence for Measure: If $(E_n)_{n\in\mathbb{N}}$ *Lebesgue Measurable*.

1. **Increasing** ↑: If $E_n \subseteq E_{n+1}$ $\Rightarrow$ $m(\cup_{n=1}^{\infty} E_n) = \lim_{n\to\infty} m(E_n)$
2. $^{\oplus}$ **decreasing** ↓: If $E_{n+1} \subseteq E_n, m(E_1) < \infty$ $\Rightarrow$ $m(\cap_{n=1}^{\infty} E_n) = \lim_{n\to\infty} m(E_n)$

   (Hint: $F_n = E_1 \setminus E_n$ (↑) $m(F_n) = m(E_1) - m(E_n)$ $m(\cup F_n) = \lim m(F_n) = m(E_1) - m(\cap E_n), m(\cup F_n) = m(E_1 \setminus \cap E_n) = m(\cap E_n) + m(\cup F_n)$)

Vitali Set: [1] 存在 not Lebesgue measurable set. [2] 存在 not countability additive set.

· **Example:** $\forall$ cosets $B \in \{x + \mathbb{Q} | x \in \mathbb{R}\}$, choose a number $x_B \in B \cap [0, 1]$ $A := \{x_B : x_B \in B\}$ [1] A is not Lebesgue Measurable Set. [2] $E := \cup_{q\in\mathbb{Q}\cap[-1,1]}(q + A)$ not Countability Additive.

8 Lebesgue Integral

8.1 Step Function || Characteristic Function || Definition of Integral

Characteristic Function: $\chi_E(x) := \{1, x \in E \;;\; 0, x \notin E \;;\; E \subset \mathbb{R}$

Basic Integral Definitions: $\int_{\mathbb{R}} \chi_I(x)\, dx = m(I) = \lambda(I)$ || $\int_{\mathbb{R}} c\chi_I(x)\, dx = c\lambda(I)$ || $\int_{\mathbb{R}} \sum c_j \chi_{I_j}(x) dx = \sum c_j \lambda(I_j)$, $c_j \geq 0$

Step Function: $f : \mathbb{R} \to [0, \infty)$ is a (non-negative) step function if $f(x) = \sum c_j \chi_{I_j}(x)$ for some c_j and χ_{I_j}

· **Property**: If f is a step function $\Rightarrow$ $\exists c_n, I_n$ s.t. I_n *disjoint* and $f(x) = \sum c_n \chi_{I_n}(x)$

Defined of $\int f dx$: $\int_{\mathbb{R}} f$ is defined iff: It's Lebesgue integrable or $f : \mathbb{R} \to [0, \infty)$ is Lebesuge Measurable.

8.2 Simple Functions || Measurable Functions

8.2.1 Simple Functions & Measurable Function

Simple Function: $s : \mathbb{R} \to \mathbb{R}$ is a simple function iff: $s(x) = \sum_{j=1}^{k} c_j \chi_{E_j}$, E_j are Lebesgue Measurable sets. || E_j can be disjoint.

Integral of Simple Function|Positive: $\int_{\mathbb{R}} s(x) dx := \sum_{j=1}^{k} c_j m(E_j)$ || $\int_{\mathbb{R}} \alpha s_1 + \beta s_2 = \alpha \int_{\mathbb{R}} s_1 + \beta \int_{\mathbb{R}} s_2$

Measurable Function: $f : \mathbb{R} \to \mathbb{R}$ is *Lebesgue Measurable* if $\forall \lambda \in \mathbb{R}, \{x \in \mathbb{R} : f(x) > \lambda\}$(**Superlevel Set**) is Lebesgue Measurable.

1. **MF|Simple|Continuous Function**: (Simple|Continuous) Function is Lebesgue Measurable. ↓Hint: $\{x | f(x) < \lambda\}^C = \cap\{x | f(x) > \lambda - \frac{1}{n}\}$
2. $^{\ominus}$ **MF|Monotone**: All Monotone functions are Lebesgue Measurable. Hint: Prove it by using definition directly.
3. **MF|Measurable Set**: If f is Lebesgue Measurable, $\Rightarrow$ $\{x \in \mathbb{R} : f(x) > \lambda \; (< \lambda \; or = \lambda)\}$ is Lebesgue Measurable.
4. **MF|property**: If f, g, f_n is Lebesgue Measurable, $\Rightarrow$ $max\{f, g\}, min\{f, g\}, f \cdot \chi_E, \alpha f \pm \beta g, f^2, f \cdot g$, $\sup f_n$, $\inf f_n$, $\limsup f_n$, $\liminf f_n$ also.
5. **Measurable Func ⇒ ∃ Simple Func**: f is 非负勒贝格可测函数 $\Rightarrow$ $\exists$ simple functions $(s_n)_{n\in\mathbb{N}}$, $0 \leq s_n \leq s_{n+1}$(↑) s.t. $s_n \to f$ pointwise.

8.2.2 Measurable Functions|Positive

Integral of Measurable Function|Positive: Let $f : \mathbb{R} \to [0, \infty)$ be Lebesgue Measurable Function.

1. **Basic Def**: $\int_{\mathbb{R}} f(x) dx := \sup\{\int_{\mathbb{R}} s(x) dx : s \; simple, 0 \leq s \leq f\}$
2. **Extend Def**: If $f : \mathbb{R} \to \mathbb{R}$ Lebesgue Measurable , E Measurable Set. $\Rightarrow$ $\int_E f(x) dx := \int_{\mathbb{R}} f(x) \chi_E(x) dx$

Basic Properties of Integral|Positive: Let $f, g : [0, \infty) \to \mathbb{R}$ be non-negative Lebesgue Measurable functions.

1. **Monotonicity**: If $0 \leq f(x) \leq g(x)$ $\Rightarrow$ $0 \leq \int_{\mathbb{R}} f(x) dx \leq \int_{\mathbb{R}} g(x) dx$
2. **Markov's Inequality**: For all $\lambda > 0$ $\Rightarrow$ $m(\{x \in \mathbb{R} : f(x) > \lambda\}) \leq \frac{1}{\lambda} \int_{\mathbb{R}} f(x) dx$ 函数积分面积 > 其对应底下矩形面积
3. $^{\oplus}$ **Measure Zero**: If E s.t. $m(E) = 0$ $\Rightarrow$ $\int_E f(x) dx = 0$ Hint: $\forall$ simple s , $0 \leq \int_E s = \int_{\mathbb{R}} s\chi_E = \int_{\mathbb{R}} c_n \chi_{E_n \cap E} = \sum c_n m(E_n \cap E) \leq \sum \cdots$

MCT for Integral|Positive: [1] $f_n : [0, \infty) \to \mathbb{R}$ non-decreasing (↑) MF; [2] $\lim f_n = f$ pointwise. $\Rightarrow$ [1] f is measurable; [2] $\int_{\mathbb{R}} \lim f_n dx = \lim \int_{\mathbb{R}} f_n(x) dx$

More Properties of Integral|Positive: Let $f, g : [0, \infty) \to \mathbb{R}$ be non-negative Lebesgue Measurable functions.

1. **Linear Combination**: $\alpha, \beta > 0$ $\Rightarrow$ $\int_{\mathbb{R}} \alpha f + \beta g = \alpha \int_{\mathbb{R}} f + \beta \int_{\mathbb{R}} g$
2. **Union of Disjoint**: E, F disjoint measurable sets $\Rightarrow$ $\int_{E\cup F} f = \int_E f + \int_F f$
3. **'MCT for Int Series'**: [1] f_n non-negative MF; [2] $\sum f_n = f$ pointwise $\Rightarrow$ [1] f is measurable; [2] $\int_{\mathbb{R}} \sum f_n = \sum \int_{\mathbb{R}} f_n$
4. **(Reverse) Fatou Lemma**: f_n non-negative MF $\Rightarrow$ $\int_{\mathbb{R}} \liminf f_n \leq \liminf \int_{\mathbb{R}} f_n \leq$ ($\limsup \int_{\mathbb{R}} f_n \leq \int_{\mathbb{R}} \limsup f_n$)

8.3 Integrating Real-Valued Functions

Isolated Functions: For a function f. We define that $f^+ := \max\{f, 0\}$, $f^- := -\min\{f, 0\}$ $\Rightarrow$ $f = f^+ - f^-$, $|f| = f^+ + f^-$.

· **Remark**: If $a < 0$: $(af)^+ = (-a)f^-$ and $(af)^- = -af^+$

Lebesgue Integrable: A MF $f : \mathbb{R} \to \mathbb{R}$ is Lebesgue Integrable iff $\int_{\mathbb{R}} |f(x)| dx < \infty$ $\Leftrightarrow$ $\int_{\mathbb{R}} f^+ dx < \infty$ & $\int_{\mathbb{R}} f^- dx < \infty$

1. **Def of Integral**: If $f \in \mathcal{L}^1(\mathbb{R})$ $\Rightarrow$ $\int_{\mathbb{R}} f(x) dx := \int_{\mathbb{R}} f^+(x) dx - \int_{\mathbb{R}} f^-(x) dx$
2. **Extended Def**: If $f : E \to \mathbb{R}$ with E Measurable Set s.t. $\tilde{f} := f(x)\chi_E$ is Lebesgue Integrable $\Rightarrow$ $\int_E f(x) dx := \int_{\mathbb{R}} \tilde{f}(x) dx$
3. $\mathcal{L}^1(\mathbb{R})$:= Collection of all Lebesgue Integrable functions on $\mathbb{R}$.
4. $\mathcal{L}^1(E)$:= Collection of all Lebesgue Integrable functions on E.

Property of Integral: Let $f, g \in \mathcal{L}^1(\mathbb{R})$. Then:

1. **Triangular Inequality for Integrals**: $|\int_{\mathbb{R}} f(x) dx| \leq \int_{\mathbb{R}} |f(x)| dx$
2. **Linearity of the Integral**: $\alpha f + \beta g \in \mathcal{L}^1(\mathbb{R})$ and $\int_{\mathbb{R}} \alpha f + \beta g = \alpha \int_{\mathbb{R}} f + \beta \int_{\mathbb{R}} g$ $\Rightarrow$ $\mathcal{L}^1(\mathbb{R})$ **is a vector space**
3. **Union of Disjoint**: E, F disjoint measurable sets $\Rightarrow$ $\int_{E\cup F} f = \int_E f + \int_F f$ (By Linearity of Integral)

4. **'Comparison Test'**: If [1] $f:\mathbb{R}\to\mathbb{R}$ measurable [2] $\int_{\mathbb{R}} g(x) < \infty$ [3] $|f| < g \;\Rightarrow\; f \in \mathcal{L}^1(\mathbb{R})$
5. **Remark**: Can use p-test, Comparison test ... when we want to prove a function is Lebesgue Integrable.

Dominated Convergence Theorem (DCT): [1] f_n is MF. [2] $f_n \to f$ pointwise. [3] $\exists g \in \mathcal{L}^1(\mathbb{R})\ s.t.\ |f_n| \le g \;\Rightarrow\;$ [1] $f \in \mathcal{L}^1(\mathbb{R})$ & $\lim \int_{\mathbb{R}} f_n = \int_{\mathbb{R}} f$

· **Uniform Convergence Theorem**: If [1] $\lim f_n(x) = f(x)$ *uniformly* on $E \in \mathcal{M}, m(E) < \infty$, [2] $f_n \in \mathcal{L}^1(\mathbb{R}) \;\Rightarrow\;$ [1] $f \in \mathcal{L}^1(E)$ [2] $\lim \int_E f_n(x)dx = \int_E f dx$

Fundamental Theorem of Calculus (FTC): $f:[a,b]\to\mathbb{R}$ be *continuous*. $\Rightarrow$ [1] $F(x) := \int_a^x f(y)dy$ is differentiable [2] $F'(x) = f(x)$

· **Corollary**: $f:[a,b]\to\mathbb{R}$ be *continuously differentiable*. $\Rightarrow \int_a^b f'(x)dx = f(b) - f(a)$

· **Remark**: [1] f is Lebesgue Integrable. (By Extreme value theorem) and. [2] 需要闭区间 (Closed) 否则需要结合 MCT 或 DCT 极限逼近来用.

1. **Integration-By-Parts**: If $f,g:[a,b]\to\mathbb{R}$ be *continuously differentiable*. $\Rightarrow \int_a^b fg'\,dx = [fg]_a^b - \int_a^b f'g\,dx$
2. **Integration-By-Substitution**: If $f:[c,d]\to\mathbb{R}$ *continuous* & $\phi:[a,b]\to[c,d]$ *continuous diff* $\Rightarrow \int_a^b f(\phi)\phi' dx = \int_{\phi(a)}^{\phi(b)} f(y)dy$

9 $\mathcal{L}^p$ Spaces || Hölder's Inequality || Minkowski's Inequality

10 Appendix

Inequation:

Triangular	$\lvert x+y\rvert \le \lvert x\rvert + \lvert y\rvert$	$\lvert x-y\rvert \ge \lvert x\rvert - \lvert y\rvert$	$\lvert\lvert x\rvert - \lvert y\rvert\rvert \le \lvert x \pm y\rvert$.	
AM-GM & Cauchy	$a+b \ge 2\sqrt{ab}, a>0, b>0$	$a^2+b^2 \ge 2ab$	$(\sum a_n b_n)^2 \le \sum a_n^2 \sum b_n^2$	
Bernoulli's Inequality	$(1+\delta)^\alpha \le 1+\delta\alpha, \alpha\in(0,1], \delta \ge -1$		$(1+\delta)^\alpha \ge 1+\delta\alpha, \alpha\in[1,+\infty), \delta\ge -1$	
$\ominus$ $e^x, ln(x)$	$e^x \ge 1+x \ge x$ Hint: Def/MVT	$\ln x \le x-1 \le x$ Hint: MVT	$e^x \ge 1+x+\frac{x^2}{2}+\cdots$ Hint: Def	$\frac{1}{k} \le ln(x) - ln(x-1)$ Hint: $\int_{k-1}^{k}\frac{dx}{x}$
常见变换技巧	$(1+\frac{x}{n})^n \ge 1+n\cdot\frac{x}{n} + C_n^2(\frac{x}{n})^2$ Hint:Binomial Theorem		$\lvert a_n\rvert\lvert x-c\rvert^n = \lvert a_n\rvert\rho^n \cdot \frac{\lvert x-c\rvert^n}{\rho^n}$, $\lvert x-c\rvert < \rho < R$	
$\ominus$ $sin(x), cos(x)$	$\lvert sin(x)\rvert \le \lvert x\rvert$ Hint:MVT	$1 - cos(x) < \lvert x\rvert$ Hint: MVT		
常见变换技巧 2	If $\forall\varepsilon>0\ a\le b+\varepsilon \;\Rightarrow\; a\le b$	$\forall\delta, 0<\delta<1$ and $\delta a \le b \;\Rightarrow\; a \le b$		

$\oplus$ 和差化积/积化和差: s+s=sc,s-s=cs,c+c=cc,c-c=-ss || e.g. $sinA + sinB = 2sin(\frac{A+B}{2})cos(\frac{A+B}{2})$ $sinAsinB = -\frac{1}{2}[cos(A+B) - cos(A-B)]$ Hint: 和差角公式相加减

· $sin(2x) = 2sin(x)cos(x)$ $cos(2x) = 2cos^2(x) - 1 = 1 - sin^2(x) = cos^2(x) - sin^2(x)$ $cos(x) = \sqrt{\frac{1+cos(2x)}{2}}$ $sin(x) = \sqrt{\frac{1-cos(2x)}{2}}$

Hyperbolic Func: $sinh(x) = \frac{e^x - e^{-x}}{2}$ $cosh(x) = \frac{e^x + e^{-x}}{2}$ $cosh^2(x) - sinh^2(x) = 1$ $[sinh(x)]' = cosh(x)$ $[cosh(x)]' = sinh(x)$

· $\sinh(x\pm y) = \sinh(x)\cosh(y) \pm \cosh(x)\sinh(y)$ $\cosh(x\pm y) = \cosh(x)\cosh(y) \pm \sinh(x)\sinh(y)$

Common Int: $\int tan(x)dx = -ln|cos(x)|$; $\int cot(x)dx = ln|sin(x)|$; $\int sec(x)dx = ln|sec(x)+tan(x)|$; $\int csc(x)dx = ln|csc(x) - cot(x)| = ln|tan(\frac{x}{2})|$

Common Der: $[tan(x)]' = sec^2(x)$; $[cot(x)]' = -csc^2(x)$; $[sec(x)]' = sec(x)tan(x)$; $[csc(x)]' = -csc(x)cot(x)$; $[arctan(x)]' = \frac{1}{1+x^2}$.

$1 + tan^2(x) = sec^2(x)$; $1 - csc^2(x) = cot^2(x)$; $csc^2(x) - cot^2(x) = 1$. $[arcsin(x)]' = \frac{1}{\sqrt{1-x^2}}$; $[arccos(x)]' = -\frac{1}{\sqrt{1-x^2}}$

Useful Counterexample: $(1/2)^{\frac{1}{n}} \subseteq [1/2, 1)$

Countably Approximation Tools: $\sum_{n=1}^{\infty} 2^{-n}\varepsilon = \varepsilon$

特殊集合构造技巧: $\{x\in\mathbb{R} : (f+g)(x) > \lambda\} = \cup_{t\in\mathbb{Q}}(\{x\in\mathbb{R} : f(x) > \lambda - t\} \cap \{x\in\mathbb{R} : g(x) > t\})$ || $\{x\in\mathbb{R} : \forall\varepsilon>0, \exists N>0\ s.t.\ \ldots < \varepsilon\} = \cap_{k=1}^{\infty}\cup_{N=1}^{\infty}\{x\in\mathbb{R} : \ \ldots < \frac{1}{k}\}$

$\oplus$ **Common Power Series**:

$f(x)$	Taylor	Series	R	$f(x)$	Taylor	Series	R
$\frac{1}{1-x}$	$\sum_{n=0}^{\infty} x^n$	$1+x+x^2+x^3+\ldots$	1	$\frac{1}{(1-x)^2}$	$\sum_{n=1}^{\infty} nx^{n-1}$	$1+2x+3x^2+4x^3+\ldots$	1
$\frac{2}{(1-x)^3}$	$\sum_{n=2}^{\infty} n(n-1)x^{n-2}$	$2+6x+12x^2+20x^3+\ldots$	1	e^x	$\sum_{n=0}^{\infty}\frac{x^n}{n!}$	$1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+\ldots$	∞
$\ln(1+x)$	$\sum_{n=1}^{\infty}(-1)^{n+1}\frac{x^n}{n}$	$x-\frac{x^2}{2}+\frac{x^3}{3}-\ldots$	1	$-\ln(1-x)$	$\sum_{n=1}^{\infty}\frac{x^n}{n}$	$x+\frac{x^2}{2}+\frac{x^3}{3}+\ldots$	1
$\sin x$	$\sum_{n=0}^{\infty}(-1)^n\frac{x^{2n+1}}{(2n+1)!}$	$x-\frac{x^3}{3!}+\frac{x^5}{5!}-\ldots$	∞	$\cos x$	$\sum_{n=0}^{\infty}(-1)^n\frac{x^{2n}}{(2n)!}$	$1-\frac{x^2}{2!}+\frac{x^4}{4!}-\ldots$	∞
$\arctan x$	$\sum_{n=0}^{\infty}(-1)^n\frac{x^{2n+1}}{2n+1}$	$x-\frac{x^3}{3}+\frac{x^5}{5}-\ldots$	1	$\sinh x$	$\sum_{n=0}^{\infty}\frac{x^{2n+1}}{(2n+1)!}$	$x+\frac{x^3}{3!}+\frac{x^5}{5!}+\ldots$	∞
$\cosh x$	$\sum_{n=0}^{\infty}\frac{x^{2n}}{(2n)!}$	$1+\frac{x^2}{2!}+\frac{x^4}{4!}+\ldots$	∞	$(1+x)^k$	$\sum_{n=0}^{\infty}\binom{k}{n}x^n$	$1+kx+\frac{k(k-1)x^2}{2!}+\ldots$	1
$\ln x$	$\sum_{n=1}^{\infty}(-1)^{n+1}\frac{(x-1)^n}{n}$	$(x-1)-\frac{(x-1)^2}{2}+\frac{(x-1)^3}{3}-\ldots$	$1, 0<x<2$	$\frac{1}{1+x}$	$\sum_{n=0}^{\infty}(-1)^n x^n$	$1-x+x^2-x^3+\ldots$	1

Sum of n, $\mathbf{n^2}$, $\mathbf{n^3}$: $\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$ $\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$ $\sum_{k=1}^{n} k^3 = [\frac{n(n+1)}{2}]^2$

Interchanging the Order of Summation: $\sum_{n=1}^{p}\sum_{m=1}^{q} f(n,m) = \sum_{m=1}^{q}\sum_{n=1}^{p} f(n,m)$ || $\oplus \sum_{k=0}^{n}\sum_{i=0}^{k} = \sum_{i=0}^{k}\sum_{k=i}^{n}$ Hint: 列出来看 or 画图

$\oplus$ **Summation By Parts**: $\sum_{k=1}^{n} a_k(b_{k+1} - b_k) = (a_{n+1}b_{n+1} - a_1 b_1) - \sum_{k=1}^{n}(a_{k+1} - a_k)b_{k+1}$, $\forall (a_k)_{k=1}^{n+1}, (b_k)_{k=1}^{n+1}$

$\Rightarrow \sum_{k=1}^{n} a_k b_k = a_{n+1}\sigma_{n+1} - \sum_{k=1}^{n}(a_{k+1} - a_k)\sigma_{k+1}$, $\forall (a_k)_{k=1}^{n+1}, (b_k)_{k=1}^{n+1}$ $\sigma_k := \sum_{j=1}^{k-1} b_j$

$\mathbf{q^n - 1}$: $q^n - 1 = (q-1)(1+q+q^2+\cdots+q^{n-1})$ || $\mathbf{1 \pm x^3}$: $1 \pm x^3 = (1\pm x)(1 \mp x + x^2)$